

Integrated Federated Causal Inference Toolkit: Testing Component Integration

Author: Daijiro Wachi (Independent OSS Engineer)

Code: <https://github.com/watilde/Harmonia>

Abstract

I built an integrated toolkit combining three federated causal inference components. **One claim: integration is computationally feasible (54k patients/sec, 264 bytes communication, 1.8M× reduction), but component interaction rules are heuristic and lack formal validation.** Tested on 1,130-2,709,803 synthetic patients. Module 1 (aggregation) achieved 15.5% narrower bounds, Module 2 (E-values) converged to FRI=2.15, Module 3 (diagnostics) scored 0.86-1.00.

Note: This demonstrates integration feasibility using synthetic Synthea data. Component interaction theory and real-world validation require future research.

1. What I Built

An integrated toolkit combining three independently tested modules:

Module 1: Optimal Aggregation - Combines site-level Manski bounds - Inverse-width weighting (15.5% improvement at 1k scale) - See companion paper: [research/modules/1-federated-partial-identification/](#)

Module 2: E-values (Robustness) - Quantifies sensitivity to unmeasured confounding - Federated Robustness Index (FRI) = sample-size weighted E-values - See companion paper: [research/modules/2-federated-evalues/](#)

Module 3: Diagnostics - Scores assumption quality (0-1 scale) - Three dimensions: unconfoundedness, positivity, specification - See companion paper: [research/modules/3-design-failure-aware-causal/](#)

Integration: Heuristic rules connect components (diagnostic scores → method selection, FRI → weight adjustment), but lack formal theory.

2. Test Results (Synthetic Data)

Table 1: Integrated Performance Across Scales

Scale	Patients	Module 1 Improvement	Module 2 FRI	Module 3 Score	Processing	Communication
1k	1,130	15.5% vs conservative	2.015	0.95	<1s	264 bytes
100k	235,222	0.22% vs conservative	2.147	0.97	4s	264 bytes
2.8m	2,709,803	0.22% vs conservative	2.149	0.98	50s	264 bytes

Key finding: All modules are computationally feasible. Integration overhead is minimal (~10% added to standalone modules).

Computational performance: - Throughput: 54,000 patients/second (2.8m scale) - Scaling: Linear $O(n)$ complexity - Memory: 2-3 GB per site

Communication: - Module 1: 150 bytes (bounds + aggregation) - Module 2: 58 bytes (E-values) - Module 3: 50 bytes (diagnostic scores) - Total: 264 bytes (constant across scales) - Reduction: 1k (762 $\times$), 100k (158,712 $\times$), 2.8m (1,825,758 $\times$)

3. Integration Rules (Heuristic, Unvalidated)

Proposed workflow:

1. **Compute diagnostics** (Module 3) $\rightarrow$ Get score (0-1)
2. **Branch by score:**
 - Score > 0.8 : Use Module 1 (bounds aggregation)
 - Score 0.5-0.8: Use Module 1 + Module 2 (bounds + E-values)
 - Score < 0.5 : Emphasize Module 2 (sensitivity analysis)
3. **Adjust weights** (proposed):
 - $w_k^{\text{final}} = w_k^{\text{base}} \times \psi(\text{score}_k) \times \phi(\text{FRI}_k)$
 - Down-weight sites with low diagnostic scores or low FRI

Critical caveat: These rules are **heuristic proposals**, not validated integration theory. They represent one possible way to connect components, not the provably correct approach.

4. Component Results

Module 1 (Aggregation): - Inverse-width: 15.5% narrower at 1k (CV=6.3%)
 - Convergence: <1% difference at 2.8m (CV=0.14%) - Communication: 150 bytes

Module 2 (E-values): - FRI: 2.015 (1k) $\rightarrow$ 2.149 (2.8m) - Interpretation: Unmeasured confounder needs RR 2.15 to nullify effect - Communication: 58 bytes

Module 3 (Diagnostics): - Scores: 0.86-1.00 (1k), 0.97-0.98 (2.8m) - All exceed exploratory 0.8 threshold - Communication: 50 bytes

5. Limitations

No integration theory: Component interaction rules are heuristic. We don't have: - Formal proof that integration preserves validity - Characterization of when to use which rule - Optimal weight adjustment formulas

Validation gap: We tested: Computational feasibility (modules run together) Individual component performance Integration benefits vs standalone use Interaction rule optimality Real-world performance under violations

Synthetic data limits: Synthea has: - Known causal structure (can't test detection of real violations) - Single data generation process (limited heterogeneity) - Simplified confounding (easier than real EHR data)

Three sites only: Real networks have 10+ sites. Interaction dynamics may differ.

Purpose: This demonstrates **integration feasibility** (can modules work together?), not **integration optimality** (is this the best way to combine them?).

6. What This Demonstrates

Modules integrate: All three can run in one pipeline

Computational feasibility: 54k patients/sec throughput

Communication efficiency: 264 bytes (constant)

Individual validation: Each module tested separately

Integration theory: No formal characterization of interaction

Comparative evaluation: Not tested vs simpler approaches

Real violations: Synthetic data can't validate detection

Optimal rules: Weight adjustment formulas are heuristic

7. Seeking Collaboration

I am an independent OSS engineer without access to real clinical data. This toolkit needs:

1. **Integration theory:** Formal analysis of component interaction

2. **Comparative studies:** Integrated vs standalone performance
3. **Real-world testing:** Multi-site hospital networks with known violations
4. **Rule calibration:** Optimize weight adjustment and branching logic

If you have expertise in causal inference theory, multi-site data, or federated systems, I welcome collaboration.

Contact: daijiro.wachi@gmail.com

8. Reproducibility

```
# Clone repository  
git clone https://github.com/watilde/Harmonia  
cd Harmonia
```

```
# Install dependencies  
npm install
```

```
# Build packages  
npm run build
```

```
# Run integrated pipeline tests  
npm run test:integrated:1k  
npm run test:integrated:100k  
npm run test:integrated:2.8m
```

Results saved to `research/modules/4-identification-sensitivity-adaptation/experiments/results/`

9. Related Work

Federated causal inference: Li et al. (2022) FACE, Zhang et al. (2023) FLAME assume unconfoundedness. Our toolkit handles unmeasured confounding via partial identification and E-values.

Multi-method integration: Most work focuses on single methods. We demonstrate feasibility of integrating aggregation, robustness quantification, and diagnostics, though optimal integration remains open.

Component papers: - Module 1: Inverse-width aggregation (Manski bounds)
- Module 2: Federated E-values (VanderWeele & Ding 2017 extended) - Module 3: Diagnostic scoring (Stuart 2010, Petersen 2012 extended)

References

1. Rosenbaum, P. R., & Rubin, D. B. (1983). The propensity score in observational studies. *Biometrika*, 70(1), 41-55.
2. VanderWeele, T. J., & Ding, P. (2017). Sensitivity analysis: introducing the E-value. *Annals of Internal Medicine*, 167(4), 268-274.
3. Li, S., et al. (2022). Federated causal inference in heterogeneous observational data. *arXiv:2202.12367*.
4. Zhang, Y., et al. (2023). Privacy-preserving federated causal inference. *AAAI*, 37(12), 14589-14597.
5. Manski, C. F. (2003). *Partial identification of probability distributions*. Springer.

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Data: Synthea synthetic healthcare data

Ethics: No IRB required (synthetic data only)